Testing Digital Signal Processing Code without an Oracle: Metamorphic Relations, Domain Generators and Error-Model Tolerances

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***Abstract*—Signal processing code is hard to test: the correct output of a filter, a transform or a resampler on an arbitrary input is unknown, the inputs that break such code are specific, and floating-point results can only be compared within a tolerance. A survey of five DSP libraries shows that their tests compare with typed-in values, never generate their inputs, and use hand-written tolerances spread over 31 orders of magnitude. We propose a method that derives the test from a statement of what an operator is: metamorphic relations proved for transforms, linear time-invariant filters, rational resamplers and streaming processors replace the expected output; inputs are drawn from signal families known to break DSP code; tolerances follow forward error bounds instead of being chosen by hand. We evaluate it with a protocol that measures its power and its soundness. On 166 functional faults injected by mutation, it detects 90 % and the ported test suites of NumPy and SciPy 96 %; the two detect different faults and together detect all of them. Ablations on hand-written faults show that the domain generators and error models make failures appear sooner, with smaller counterexamples and no false alarm. On correct code the method raised no false alarm in 2 000 runs. Applied to SciPy, it found faults on empty blocks that we reported with a fix.**

***Index Terms*—software testing, metamorphic testing, property-based testing, digital signal processing, floating-point error, test oracle**

I. Introduction

Digital signal processing (DSP) code sits at the bottom of a large software stack: the fast Fourier transform (FFT), linear filters and sample-rate converters of NumPy [1], SciPy [2] and PyTorch [3] are called by audio, communication, geophysics and machine-learning code that rarely re-checks them. A fault at this level propagates silently: a filter that loses its state on an empty block does not crash the application that streams audio through it, it makes the output diverge. Testing these operators faces three problems at once.

P1, no oracle: For most inputs nobody knows the exact output of a filter or a transform. This is the *oracle problem* of software testing [4], and it is acute for scientific software [5]. In practice, tests compare against values computed by another tool and typed into the test file, or against a second implementation that may share the same assumptions. In the suites we surveyed (Section III), 169 tests of `scipy.signal` compare with hard-coded arrays.

P2, inputs that do not exercise the faults: Hand-written tests use a few lengths and smooth or random signals. Faults in DSP code, however, hide in specific places: lengths that are prime, odd or one away from a power of two (which change the FFT algorithm), tones between two DFT bins or close to the Nyquist frequency, extreme amplitudes, and, for streaming code, empty or one-sample blocks. A concrete case: `scipy.signal.lfilter` returned an uninitialised final state when given an empty block and an initial state (SciPy issue 22571[1]), so that any block-processing loop silently diverged after an empty block, and `sosfilt` still raises on such input. Across the 2 388 tests we surveyed, only 2 combine an initial state with an empty input, and none generates its inputs.

P3, no principled tolerance: Even when the expected output is known, floating-point results are only close to it, and *how close* depends on the operation, the length and the conditioning of the problem. Tests therefore use tolerances chosen by hand: 66 distinct values in `scipy.signal`, from 10^{-31} to 3.01. A tolerance that is too loose hides a single-precision intermediate; one that is too tight produces false alarms, which teach maintainers to loosen tolerances further.

Property-based testing [6], [7] answers P2 in general by generating inputs, and metamorphic testing [8], [9] answers P1 in general by checking relations between outputs instead of the outputs themselves. Neither tells a DSP developer which relations hold for which operator, which signals to generate, nor how to compare floating-point results. And a method that claims to find faults in numerical libraries must itself be shown not to raise false alarms, since a single spurious report costs a maintainer's trust.

This article answers the three problems with one method and evaluates it with a protocol designed to measure both its power and its soundness. Its contributions are, in order of importance:

- *A method* that replaces the expected output by metamorphic relations proved for four classes of DSP operators (transforms, linear time-invariant filters, rational resamplers, streaming processors), draws inputs from the signal families that break DSP code, and derives each tolerance from a forward error bound instead of choosing it (Section IV).

[1] https://github.com/scipy/scipy/issues/22571

- *An evaluation protocol* that compares the method with the test suites of NumPy and SciPy on automatically generated faults, isolates the contribution of the generators and of the tolerances by ablation, and validates the method itself: false alarms on correct code, margin between observed error and tolerance, accuracy of the references against 50-digit arithmetic, and an audit of the generators (Section V).
- *Faults found in widely used libraries* and reported to their maintainers, with a proposed fix (Section VI-D).
- *A survey* of how five DSP libraries test themselves, which grounds P1–P3 in counts rather than impressions (Section III).

The method is implemented in dsp-props, an open-source layer on top of Hypothesis; the library, the experiments and their data are available at https://github.com/clement-basdevant/dsp-props under a BSD licence.

Section II reviews related work, Section III reports the survey, Section IV describes the method, Section V the evaluation protocol and Section VI its results. Section VII discusses what the results mean for practitioners, and Section VIII concludes.

II. Background and related work

A. The oracle problem and metamorphic testing

A test oracle decides whether an output is correct. When no such procedure exists, or when it is too costly, testing faces the oracle problem [4]; scientific software is the textbook case, since it is often written precisely because the answer is not known in advance [5]. Metamorphic testing [8], [9] avoids the oracle by checking *metamorphic relations*: if f is correct, then for related inputs x and x' the outputs $f(x)$ and $f(x')$ must satisfy a known relation, e.g., $f(x + x') = f(x) + f(x')$ for a linear f. It has been applied among others to bioinformatics [10] and image processing [11]. The quality of the approach depends on the relations: a relation that holds for the faulty program too detects nothing, and finding relations is the main cost of the method. For DSP operators the relations are classical theorems (linearity, shift and convolution theorems, Parseval's identity), but which ones apply depends on what the operator is, and on finite sequences their exact form at the edges matters.

B. Property-based testing

Property-based testing (PBT), introduced by QuickCheck [6] and available in Python through Hypothesis [7], states a property over all inputs of a type, generates inputs and, when one fails, *shrinks* it to a smaller, simpler failing input [12]. PBT supplies the machinery to check metamorphic relations, but not the relations themselves, nor the generators of meaningful inputs. Interviews of PBT users identify writing properties and generators as their main difficulty [13], and Maaz et al. use language models to propose properties for Python libraries [14]. Generic generators of floating-point arrays also produce values, such as subnormal numbers, for which a purely relative tolerance is meaningless; Section VI-B shows a false alarm of this kind.

C. Self-testing of numerical programs

Blum, Luby and Rubinfeld [15] showed that a program computing a linear function can be tested, and even corrected, by checking random linear identities. Ergün [16], [17] extended the idea to multivariate linear functions such as the DFT, and Ergün, Kumar and Rubinfeld [18] to computations that are only approximately correct. The FFTW library [19] ships a self-test that implements Ergün's tester and checks linearity, impulse responses and time and frequency shifts [20]. These works establish that linearity and shift identities characterise the DFT; they do not cover filters, resamplers or streaming code, and they treat the floating-point error with a single threshold.

D. Differential testing and numerical faults

Differential testing compares several implementations of the same function [21]. It needs at least two of them, and they may share an algorithm or an assumption, e.g., the same handling of an empty block. Di Franco et al. studied 269 numerical bugs reported in NumPy, SciPy, LAPACK, GSL and Elemental [22]. They selected bugs confirmed by developers, most of them closed with a fix, and analysed 183 of those fixes: the study shows that numerical faults reach widely used, mature libraries, not that these particular faults remain in them.

E. Floating-point error of DSP operators

The FFT computed by a Cooley–Tukey algorithm has a worst-case relative error bounded by $O(\varepsilon \log N)$ [23], where ε is the unit roundoff, and its root-mean-square error on random inputs grows only as $O(\varepsilon\sqrt{\log N})$ [24]; a direct DFT, computed as N dot products of length N, has a bound that grows as $O(\varepsilon N)$ [23]. For a linear recursion, rounding errors are fed back through the poles: Appendix B derives from the standard model of floating-point arithmetic [23] a bound that grows with the ℓ_1 norm of the impulse response, which is large when poles approach the unit circle. These bounds are rarely used in tests, which fall back on fixed relative tolerances; we use them to set tolerances (Section IV-D).

F. Mutation analysis

Mutation analysis [25], [26] measures the fault-detection power of a test approach by injecting small syntactic changes, *mutants*, into a correct program and counting how many the tests detect (*kill*). Mutation operators replace an arithmetic, relational or bitwise operator, change a constant, delete a statement, and so on. Offutt et al. [27] showed that a small selective set of operators (arithmetic, relational and logical operator replacement, operand replacement and unary operator insertion) yields almost the same mutation score as the full set, which keeps mutation affordable. Two assumptions underpin the method: the *competent programmer* hypothesis, under which real faults are small deviations from a correct program, and the *coupling effect*, under which tests that kill simple faults

TABLE I
WHAT EACH APPROACH SUPPLIES TO THE DEVELOPER: P1, NO NEED FOR EXPECTED VALUES; P2, GENERATED INPUTS; P3, A DERIVED TOLERANCE

Approach	P1	P2	P3
Expected-value tests	–	–	–
Differential [21]	needs 2nd impl.	partly	–
DFT self-testing [16], [20]	DFT only	yes	–
Generic PBT, MT [7], [8]	user's relations	generic	–
This work	yes	DSP families	yes

also kill complex ones [25]. Just et al. found that mutant detection correlates with the detection of real faults in Java programs [28].

Two difficulties matter here. First, some mutants are *equivalent*: they compute the same function as the original, and no test can kill them; deciding equivalence is undecidable in general [25], so studies use heuristics. Numerical code adds a third category: a mutant may change the output only by rounding, e.g., by reordering an addition. Such a mutant is neither equivalent nor functionally wrong, and whether it should be killed depends on the accuracy the program promises. We classify mutants with an independent oracle into equivalent, accuracy and functional faults (Section V-C) and report kill rates per class, so that a false alarm on an equivalent mutant counts against an approach instead of for it. Second, first-order mutants are mostly easy to kill; we therefore also use hand-written faults modelled on real DSP bugs, including accuracy faults.

G. Summary

Table I summarises what each family of approaches leaves to the developer. Every approach answers part of P1–P3; none supplies relations, generators and tolerances together for the operators of a DSP library, which is the gap this article addresses.

III. HOW DSP LIBRARIES TEST THEMSELVES

To ground P1–P3 in data, we analysed the test suites of `numpy.fft`, `scipy.fft`, `scipy.signal`, librosa [29] and the functional module of torchaudio, 2 388 test functions in total.

Method: A script parses every test file into a syntax tree and counts, per library: the test functions and those decorated with Hypothesis' `@given`; the numerical comparisons, i.e., calls to `assert_allclose`, `assert_array_almost_equal`, `allclose`, `torch.testing.assert_close` and their relatives; among them, those with a hand-written tolerance, i.e., a literal passed as `rtol`, `atol`, `decimal`, `nulp` or similar, converted to a magnitude (`decimal` $= d$ means 1.5×10^{-d}); the tests that pass an empty signal, an initial filter state (`zi=`), or both; and the tests that compare with a typed-in array of at least four floating-point literals. These are syntactic heuristics: a tolerance computed at run time or an expected array loaded from a file is not counted, so the counts are lower bounds.

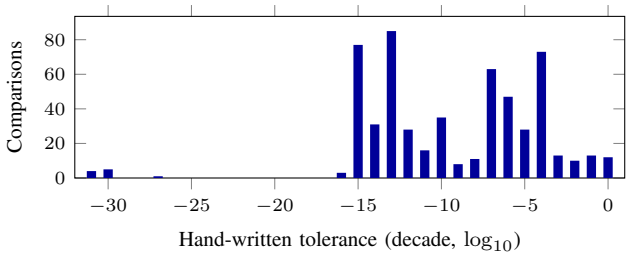


Fig. 1. Hand-written tolerances in the tests of `scipy.signal`, grouped by decade.

Three typical tests: The ported suites of Section V give a concrete picture. SciPy's `test_rank_1_IIR` filters the ramp $0, 1, \ldots, 5$ with $b = [1, -1]$, $a = [0.5, -0.5]$ and compares the output with the six typed-in values $0, 2, 4, 6, 8, 10$ to six decimals: one input, one filter, one tolerance. `test_length_factors` for `upfirdn` checks five combinations of lengths and rates against output vectors whose comment says they come from MATLAB. `test_vs_convolve` compares `upfirdn` with a straightforward implementation on 10 000 random samples, for three decimation factors, at a relative and absolute tolerance of 10^{-7}, which is a form of differential testing within the library. Several tests are regression tests added after an issue was reported, such as the two `test_short_x` tests for a one-sample input with an initial state: edge cases enter the suites after a user has met them.

No oracle (P1): Table II shows that hard-coded expected arrays are common: 220 tests in total, of which 169 in `scipy.signal`. Such arrays come from another tool, MATLAB according to the comments of some tests, or from an earlier run of the library, and they test the library against that source on a handful of inputs. The other tests compare two computations of the same quantity within the library, e.g., `lfilter` against `convolve`, which is a form of differential testing.

Few, fixed inputs (P2): No test in the five suites uses Hypothesis or any other input generator. In the tests we ported, inputs are written by hand: short ramps, a few random arrays with a fixed seed, constant signals. Edge cases are present but scattered: `scipy.signal` has 62 tests with an empty input and 19 with an initial state, but only 2 that combine both, which is the case behind the faults of Section VI-D.

Hand-written tolerances (P3): `scipy.signal` alone performs 1 645 comparisons, 584 of them with a hand-written tolerance, which take 66 distinct values spanning 31 orders of magnitude (Fig. 1). The most frequent are 10^{-13} (85), 10^{-15} (77), 10^{-4} (73). The spread does not follow the operations: tolerances near 10^{-15} assert agreement to the last bits, tolerances near 10^{-4} would accept a single-precision computation, and a few tolerances above 1 accept almost anything. Each value is what made a test pass on its author's machine, which is P3 in its plainest form.

TABLE II

TEST SUITES OF FIVE DSP LIBRARIES (SYNTACTIC COUNTS). TOL.: COMPARISONS WITH A HAND-WRITTEN TOLERANCE; DISTINCT: NUMBER OF DISTINCT TOLERANCE VALUES; EMPTY: TESTS WITH AN EMPTY INPUT; ZI+EMPTY: TESTS THAT ALSO PASS AN INITIAL STATE; HARD-CODED: TESTS THAT COMPARE WITH AN ARRAY OF AT LEAST FOUR TYPED-IN FLOATING-POINT VALUES

Suite	Tests	Hypothesis	Comparisons	Tol.	Distinct	Range	Empty	zi+empty	Hard-coded
NumPy (numpy.fft)	52	0	102	66	3	10^{-14}–10^{-6}	1	0	1
SciPy (scipy.fft)	96	0	132	11	7	10^{-10}–3.0×10^{-6}	0	0	4
SciPy (scipy.signal)	1095	0	1645	584	66	10^{-31}–3.01	62	2	169
librosa	916	0	664	60	20	10^{-12}–3	4	0	36
torchaudio (functional)	229	0	13	12	12	10^{-7}–0.1	0	0	10

IV. METHOD

The method rests on one idea: the user states *what an operator is*, and everything else follows from that statement. A DFT, a causal LTI filter with a given transfer function, an L/M resampler or a streaming processor each satisfy known theorems (the relations), are known to fail on known inputs (the generators), and have a known forward error (the tolerance). The method is implemented in dsp-props as a layer on top of Hypothesis, but its ingredients do not depend on that implementation.

A. Declarations

A declaration is a decorator on the operator under test. For instance,

```
@dp.lti_filter(transfer=(b, a))
def my_filter(x):
    return scipy.signal.lfilter(b, a, x)

dp.verify(my_filter)
```

declares a causal linear time-invariant (LTI) filter with transfer function $B(z)/A(z)$. `dp.verify` runs every relation that applies to the declared class on generated inputs (200 per relation by default) and raises an error that carries the smallest failing input found by shrinking, the relation that failed, the observed error and the tolerance. The declarations available are transforms (complex DFT, DFT of real input, generic linear transforms such as the DCT), LTI filters (with an optional transfer function and gain bound), rational resamplers, and streaming or stateful processors; adapters wrap NumPy, SciPy, PyTorch and C functions called through `ctypes`.

A declaration fixes the parameters of the operator: the coefficients of a filter, the rates of a resampler. The relations then vary the input signal only. This is a deliberate choice, since parameters such as filter coefficients must satisfy constraints (stability, a declared transfer function) that a generic generator cannot know; Section VI-A shows its cost.

B. Relations (answer to P1)

Table III lists the relations for each operator class. Each is a consequence of the declaration [30], [31] and is proved in Appendix A; we state them on finite sequences of length N, as implementations compute them, which matters for shifts and edges. For a transform, the shift theorem is the *circular* one, and the shift is reduced modulo N in integers so that the twiddle factors of the reference are themselves accurate. For a filter, a delay is applied by prepending zeros, and the delayed output is compared with the original output, also delayed, from zero state. For a resampler, a delay of M input samples must delay the output by exactly L samples, and the output length must be $\lceil NL/M \rceil$. For a streaming processor, the input is cut into blocks at generated positions, including empty and one-sample blocks, and the concatenated outputs must equal the one-shot output.

The relations replace the oracle: no expected output is needed, except for the optional transfer function, which the user states once instead of typing expected arrays. Without it, an LTI filter is only checked to be linear, time-invariant and causal, so a filter with the wrong coefficients but the right structure passes; this blind spot is measured in Section VI-A.

C. Generators (answer to P2)

Signals are drawn from seven families, each aimed at a known class of fault: impulses (indexing and edge errors), constants (DC gain, normalisation), sinusoids at arbitrary frequency, including off-grid ones (leakage and twiddle errors), sinusoids near the Nyquist frequency (aliasing, sign alternation), pairs of tones less than one DFT bin apart (resolution, accumulated rounding), sparse signals and white noise. Amplitudes are drawn from $\{10^{-100}, 10^{-3}, 1, 10^{3}, 10^{100}\}$ to reach underflow and overflow paths. Lengths are biased towards the values that change the code path of an FFT or a filter: 1, 2, small primes, and $2^k - 1$, 2^k, $2^k + 1$, mixed with uniform lengths up to 512. Block splits for streaming code include empty and one-sample blocks.

The families are ordered so that Hypothesis' shrinking converges to short impulses of unit amplitude, which yields counterexamples a DSP engineer can read at a glance: on the hand-written faults of Section VI-B, the median counterexample has 2 samples.

D. Tolerances (answer to P3)

All comparisons use the ℓ_2 norm of the error, scaled by the norm of a reference quantity, and a model of the admissible error. Using norms rather than element-wise comparisons follows the form of the error bounds, which are normwise, and avoids spurious failures on output samples close to zero.

FFT. For transforms,

$$\|\hat{y} - y\|_2 \le c\,\varepsilon\, \max(\log_2 N, 1)\, \|y\|_2 + N \,\mathrm{tiny}, \qquad (1)$$

TABLE III
RELATIONS CHECKED FOR EACH OPERATOR CLASS. T: TRANSFORM; H: FILTER; R: RESAMPLER; D_s: DELAY BY s SAMPLES WITH ZERO FILLING; S_s: CIRCULAR SHIFT; $\omega = e^{-2\pi i/N}$

Class	Relation	Checked property
Transform	Linearity	$T(\alpha x + \beta y) = \alpha Tx + \beta Ty$
	Circular shift theorem	$T(S_s x)[k] = \omega^{ks}\, Tx[k]$
	Parseval's identity	$\|Tx\|_2^2 = c\,\|x\|_2^2$, with c declared
	Hermitian symmetry	x real $\Rightarrow Tx[N-k] = \overline{Tx[k]}$; DC and Nyquist bins real
	Inversion	$T^{-1}(Tx) = x$
	Convolution theorem	$T(x \circledast y) = Tx \cdot Ty$
LTI filter	Linearity	$H(\alpha x + \beta y) = \alpha Hx + \beta Hy$, from zero state
	Time invariance	$H(D_s x) = D_s(Hx)$
	Causality	changing $x[n]$ for $n \geq k$ leaves $y[n]$ unchanged for $n < k$
	Frequency response	DFT of the impulse response equals $B(e^{i\theta})/A(e^{i\theta})$ on the DFT grid
	Bounded gain	$\|x[n]\| \leq 1$ for all $n \Rightarrow \|y[n]\| \leq G$, with G declared
Resampler L/M	Linearity	as for filters
	Output length	$\|Rx\| = \lceil NL/M \rceil$
	Periodic time invariance	$R(D_{Ms}x) = D_{Ls}(Rx)$
	DC gain	a constant input gives the declared constant away from the edges
	Alias rejection	a tone above the output Nyquist frequency is attenuated, not folded back
Streaming	Block invariance	any split into blocks, including empty and one-sample blocks, gives the one-shot output

the form of the forward error bound of the radix-2 FFT [23, Thm. 24.2], with a constant c that covers the few chained transforms in a relation, and a floor at N times the smallest normal number tiny. The floor keeps exact zeros strict while tolerating underflow on signals of amplitude 10^{-100} or subnormal values, where a purely relative tolerance would be zero.

Linear recursions. For a filter $y = (B/A)x$ computed by a direct-form recursion, Appendix B derives

$$\|\hat{y} - y\|_2 \leq c\,\varepsilon\,k\,\|h_A\|_1\big(\|b\|_1 + \|a_{1:}\|_1\|h_{B/A}\|_1\big)\|x\|_2, \quad (2)$$

to first order in ε, where k is the number of coefficients other than $a_0 = 1$, h_A and $h_{B/A}$ are the impulse responses of $1/A$ and B/A, and $a_{1:}$ are the feedback coefficients. The bound grows as poles approach the unit circle, which is where recursions lose accuracy, and reduces to the error of a dot product for a FIR filter.

Calibration. An analytic bound cannot describe every library: an FFT may switch algorithm with the factorisation of N, and its error then varies by two orders of magnitude between neighbouring lengths (Fig. 2). `dp.calibrate` measures the error of an implementation against an extended-precision reference for every length, over several random inputs, and turns the worst error times a margin of 8 into a per-length tolerance. It refuses to calibrate when the error exceeds what rounding can explain ($10^6\varepsilon$), so that a fault cannot be absorbed into the tolerance.

E. Differences with the state of the art

Compared with generic PBT, the method supplies the three pieces of domain knowledge that PBT leaves to the user; Section VI-B measures the contribution of two of them. Compared with self-testing of the DFT [16], [20], it covers filters, resamplers and streaming code, and its tolerance follows the operator. Compared with differential testing [21], it needs a single implementation. Compared with the test suites of Section III, it needs no expected values and no hand-chosen tolerance.

V. EVALUATION PROTOCOL

We answer four research questions (RQ). **RQ1**: does the method detect more faults than the state of practice? **RQ2**: what do the domain generators and the error-model tolerances contribute? **RQ3**: is the method itself correct, i.e., free of false alarms, with sound references and generators? **RQ4**: does it find faults in real libraries? RQ1 and RQ2 use injected faults, RQ3 correct code, RQ4 the libraries themselves.

A. Subjects

We wrote four small reference implementations with the interface of a library function: a radix-2 FFT with Bluestein's algorithm for other lengths, a direct-form II transposed `lfilter` with initial state, a polyphase `upfirdn` used by a port of `resample_poly`, and a block FIR filter by overlap-add. They are short enough for exhaustive mutation and use the same algorithms as the libraries they imitate. All four agree with NumPy or SciPy to a relative error below 1.6×10^{-15} and pass every suite below.

B. Faults

A mutation tool generates every first-order mutant of each subject with standard operators, following [25], [27]: arithmetic, bitwise and relational operator replacement, constant replacement ($k \to k \pm 1$, $c \to c + 1$), unary minus deletion, `min`/`max` swap and statement deletion. This gives 183 mutants (76 FFT, 55 `lfilter`, 36 `upfirdn`, 16 overlap-add).

We also use 31 hand-written faults in six families, modelled on real DSP bugs: a forgotten $1/N$, a twiddle sign error, single-precision intermediates, a filter that changes after 64 samples, a gain wrong by 1 %, a missing interpolation gain, a state reset between blocks. Four of them are accuracy faults (single-precision intermediates), which first-order mutation does not produce.

C. Ground truth

An oracle used only for classification runs each mutant on a fixed battery of inputs and compares it with an independent reference: all lengths up to 130 and several larger ones for the FFT; many filters, including order 0, and initial states for `lfilter`; rates up to $5/4$ and $1/7$ for `upfirdn`, with and without default arguments; random block splits for overlap-add. A mutant is *equivalent* if its largest relative error is at most 10^{-12}, an *accuracy* fault if it is between 10^{-12} and 10^{-6}, and a *functional* fault above 10^{-6}, or if it raises, returns the wrong shape or exceeds the time limit. This gives 166 functional faults and 17 equivalent mutants. No mutant falls in the accuracy band: the largest relative error of a mutant is either below 10^{-14} or above 1. First-order mutation of these subjects thus produces large faults only, and the hand-written faults cover accuracy.

D. Approaches compared

SoP (state of practice) is the test suite of the library whose interface the subject implements, ported test by test with its inputs and tolerances: NumPy's and SciPy's FFT tests, SciPy's `lfilter`, `lfilter_zi`, `upfirdn` and `resample_poly` tests, including the regression test added for issue 22571. The overlap-add subject, which has no library counterpart, gets four tests in the same style. *dsp-props* runs the declarations a user would write, two to four per subject, written before the mutants were run. Three ablations isolate the components: *uniform* replaces the domain generators by uniform lengths in $[1, 512]$ and values in $[-1, 1]$; *fixed-tol* replaces every tolerance model by a fixed relative tolerance of 10^{-7}, the default of `numpy.testing.assert_allclose`; *generic PBT* applies both, i.e., the same relations written with Hypothesis alone.

E. Metrics and execution

For each approach and mutant we record whether the mutant is detected, the number of generated inputs before the first failure, the length of the shrunk counterexample and the run time. Each randomised approach runs with 5 seeds and we report the mean, with the range across seeds when they disagree; SoP is deterministic and runs once. A run that exceeds 45 s counts as a detection, since a hanging test fails in continuous integration. All runs use 200 inputs per relation, the library default, in the following environment: Python 3.13.16, NumPy 2.5.4, SciPy 1.18.1, Hypothesis 6.169.3, on a 4-core Linux x86-64 machine.

VI. Results

A. RQ1: detection compared with the state of practice

Table IV gives the share of mutants detected, averaged over seeds. dsp-props detects 90 % of the 166 functional mutants and the ported suites 96 %; no approach raises an alarm on an equivalent mutant. The two sets of detected faults are not nested: 17 mutants are detected only by the suites, 6 only by dsp-props, and together they detect 100 % of them.

Every miss of dsp-props has the same cause: the fault shows only for operator parameters that no declaration uses. A declaration fixes the coefficients of a filter or the rates of a resampler, and dsp-props then varies the input signal. Seven `lfilter` mutants break filters of order 0 or 1, or filters whose numerator is shorter than the denominator, and the declared filters have orders 4 and 2 with numerators at least as long. The `upfirdn` survivors change the length of the full output, which the declarations never see since they reach `upfirdn` through `resample_poly` and through a FIR filter truncated to the input length, or ignore the last input sample, which every relation sees identically on both of its sides. The suites miss the opposite kind of fault. Their only comparisons with expected values on the FFT use length 30, which the subject computes with Bluestein's algorithm; on powers of two they only check that `ifft` inverts `fft`, which two mutants of the radix-2 twiddle factors still satisfy. They also always pass the rates of `upfirdn` explicitly, so four mutants of its default arguments survive.

dsp-props therefore does not detect more of these faults than suites refined over many years; it detects other ones, without expected values, and its misses are a matter of declarations rather than of relations: declaring a first-order filter would have caught the `lfilter` survivors.

B. RQ2: contribution of generators and tolerances

On the generated mutants the four randomised variants detect the same faults, up to one FFT mutant that the variants with uniform inputs miss in some seeds (Table IV). These mutants change the output by orders of magnitude, and almost any input reveals them: the median number of inputs before the first failure is 3 for dsp-props and 2 for uniform inputs, and the median counterexample has 3 and 2 samples. The generators and tolerances do not change what is detected here, and the generators cost time, with a median of 3.21 s per mutant against 1.6 s for uniform inputs and 0.001 s for a ported suite.

The hand-written faults, which include the accuracy faults, separate the variants (Table V). Every variant detects every fault in every seed, but not equally fast. With the domain generators, the median failure appears at input 3 (90th percentile 7); with uniform inputs at input 8 (22), and with generic PBT at input 8 (39). Counterexamples are smaller too: the 90th percentile of their length is 17 samples for dsp-props, 34 with uniform inputs, and 97 with a fixed tolerance. Finally, generic PBT raised a false alarm on the correct real FFT, in 1 of its 5 runs: Hypothesis drew a subnormal input of norm 10^{-323},

TABLE IV

MUTANTS DETECTED (%, MEAN OVER SEEDS, WITH MIN–MAX WHEN SEEDS DISAGREE). FOR EQUIVALENT MUTANTS THE FIGURE IS A FALSE-ALARM RATE AND SHOULD BE 0. SOP: PORTED TEST SUITES OF NUMPY AND SCIPY; UNIFORM: DSP-PROPS WITHOUT ITS GENERATORS; FIXED-TOL: WITHOUT ITS TOLERANCE MODELS; GENERIC PBT: WITHOUT BOTH

Subject	Class (mutants)	SoP	dsp-props	uniform	fixed-tol	generic PBT
FFT	functional (63)	97	100	100 (98–100)	100	100 (98–100)
FFT	equivalent (13)	0	0	0	0	0
`lfilter`	functional (51)	100	86	86	86	86
`lfilter`	equivalent (4)	0	0	0	0	0
`upfirdn`	functional (36)	89	72	72	72	72
overlap-add	functional (16)	100	100	100	100	100
all	functional (166)	96	90	90 (89–90)	90	90 (89–90)
all	equivalent (17)	0	0	0	0	0

TABLE V

HAND-WRITTEN MUTANTS BY FAMILY: RUNS THAT REPORT A FAILURE, OVER MUTANTS AND 5 SEEDS. A CORRECT REFERENCE MUST NEVER FAIL

Mutant	dsp-props	uniform	fixed-tol	generic PBT
FFT, reference	0/5	0/5	0/5	0/5
FFT, 8 mutants	40/40	40/40	40/40	40/40
real FFT, reference	0/5	0/5	0/5	1/5
real FFT, 5 mutants	25/25	25/25	25/25	25/25
DCT, reference	0/5	0/5	0/5	0/5
DCT, 3 mutants	15/15	15/15	15/15	15/15
IIR filter, reference	0/5	0/5	0/5	0/5
IIR filter, 5 mutants	25/25	25/25	25/25	25/25
resampler, reference	0/5	0/5	0/5	0/5
resampler, 6 mutants	30/30	30/30	30/30	30/30
streaming, reference	0/5	0/5	0/5	0/5
streaming, 4 mutants	20/20	20/20	20/20	20/20
mutants always detected	31	31	31	31
first failure, median (p90)	3 (7)	8 (22)	3 (10)	8 (39)
counterexample length, p90	17	34	97	35

for which a purely relative tolerance is zero and a rounding error of one subnormal fails the test. The error model of dsp-props includes an absolute floor, and it raised no false alarm. With 200 inputs per relation, these differences do not change the detection rate. We expect them to matter with a smaller budget or on faults that need a specific input, but we did not measure this. One hand-written fault, a filter whose gain is wrong by 1 %, is detected only because its transfer function is declared: it is still linear, time-invariant and causal.

C. RQ3: is the method itself correct?

A reader of any fault-finding method should ask whether its alarms are real. We check four things.

Relations: Every relation is a theorem about the declared class of operators (Appendix A), so a correct implementation in exact arithmetic never violates it; only the floating-point comparison can raise a false alarm.

False alarms: We ran every declaration on correct implementations, NumPy's FFT and real FFT, SciPy's `lfilter`, `sosfilt`, `upfirdn` and `resample_poly`, and the four reference subjects, with 20 independent seeds: 2 000 runs of a relation, 248 000 floating-point comparisons, no failures. SciPy's `lfilter` with an initial state on empty blocks is excluded from this count because it is the known fault of issue 22571; dsp-props flags it in 5 of 5 runs on SciPy 1.18.1.

Margin: During the same runs we recorded the ratio of the observed error to the tolerance for every comparison. 63 % of the comparisons are exact (ratio 0); the 99th percentile is 0.038 and the maximum 0.20: the tolerance is 5 times larger than the worst error observed on correct code. It is nevertheless tight enough to catch the four single-precision faults (Table V).

References: Calibration and the tests of the IIR model compare against extended-precision references, a DFT computed by NumPy in `long double` and a direct recursion in `long double`. Against a 50-digit computation with mpmath [32], their relative error is at most $1.0 \times 10^{-3}\,\varepsilon$ and $5.4 \times 10^{-3}\,\varepsilon$ respectively, where ε is the double-precision unit roundoff: the references are three orders of magnitude more accurate than what they are used to measure.

Generators: We sampled each generator and checked its claims. All 19 pathological lengths are produced, impulses have exactly one non-zero sample, all signals are finite and of the requested length, and amplitudes cover 10^{-100} to 10^{100}; 78 % of block splits contain an empty block and 37 % a one-sample block; near-Nyquist frequencies lie in $[0.45, 0.5]$; 99 % of the tone pairs are less than one bin apart. The audit also found a weakness: the "off-grid" sinusoid lands exactly on a DFT bin in 39 % of draws, because Hypothesis favours simple values such as 0 and 0.5 (23 % of draws). These are useful edge cases, but the family is less often off-grid than its name says; a future version will draw both on-grid and strictly off-grid frequencies.

D. RQ4: faults in real libraries

We ran declarations of 46 operators of SciPy, NumPy, PyTorch, librosa and torchaudio. The campaign produced the following findings.

- `scipy.signal.lfilter` returns an uninitialised final state, and `sosfilt` raises, when given an empty block with an initial state. The `lfilter` part is an independent rediscovery of issue 22571, fixed in the development version after SciPy 1.18.1; the `sosfilt` part, which fails on any input that is empty along the filtered axis, was new. The same fault makes librosa's streaming pre-emphasis fail on an empty block.
- `scipy.signal.tf2sos` drops pure delays: for $y[n] = x[n-2]$ it returns an identity sec-

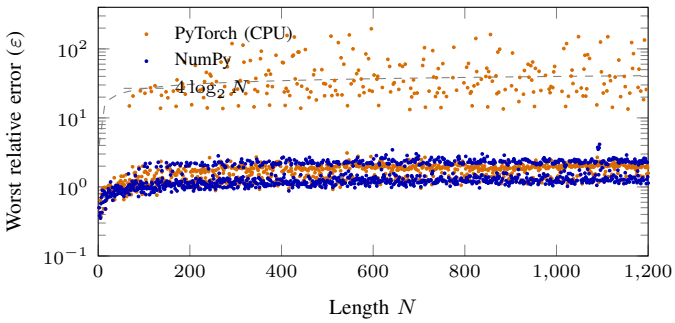


Fig. 2. Worst relative FFT error, in units of ε, versus length N, against a `long double` reference (`benchmarks/error_profile.py`). SciPy, not shown, behaves like NumPy.

tion, so `sosfilt(tf2sos(b, a), x)` differs from `lfilter(b, a, x)`.

- torchaudio's `deemphasis`, documented as the inverse of `preemphasis`, clamps its output to $[-1, 1]$.
- PyTorch's CPU FFT reaches $195\,\varepsilon$ for $N = 596$, against at most $4.1\,\varepsilon$ for NumPy over $N = 2, \ldots, 1200$ (Fig. 2); this is not a fault, but it shows why one fixed tolerance cannot fit every library.

The campaign also produced three alarms that triage traced to errors in our own declarations (e.g., `filtfilt` is not symmetric under time reversal with its default padding); they are listed in the repository.

E. Threats to validity

Construct. Mutants are a proxy for real faults [28]; the hand-written faults and the faults of RQ4 complement them. The oracle thresholds (10^{-12}, 10^{-6}) are a choice; the per-mutant errors are in the result files so that other thresholds can be applied. The oracle was extended once after the first run: it did not call `upfirdn` with its default arguments nor `lfilter` with an order-0 filter, so it labelled as equivalent eight mutants that the SoP suites or dsp-props detected; with the extended oracle, no approach detects any equivalent mutant. *Internal.* The kill rates of dsp-props depend on the declarations, which we wrote before running the mutants and did not change afterwards. Results depend on random seeds, which we vary and report. *External.* The subjects are small, and the SoP suites are ports of real suites restricted to the subject's interface; the original suites also test other data types and array shapes. The survey counts are syntactic lower bounds.

Reproducing: Every number and table of this article is produced by the following scripts, all in the `benchmarks/` folder of the repository:

- `survey_test_suites.py`: survey of Section III;
- `comparison/run_comparison.py`: RQ1 and RQ2, generated mutants;
- `comparison/run_hand_mutants.py`: RQ2, hand-written faults;
- `validation.py`: RQ3;
- `campaign.py` and `error_profile.py`: RQ4.

The script `paper/article/make_numbers.py` turns their result files into the numbers and tables of the article, and `proofs/check.sh` checks the Coq proofs of the appendices.

VII. Discussion

What the relations can and cannot see: A relation detects a fault only if the fault breaks it. Faults that change the operator into another operator of the same class, a filter with the wrong coefficients but still LTI and causal, a resampler that ignores the last input sample consistently, satisfy every structural relation. The transfer-function relation closes the first gap at the cost of one declaration; the second remains. Relations are thus strongest on faults of structure (state, edges, indexing, normalisation) and weakest on faults of value, where expected outputs remain useful.

Parameters are part of the input: The misses of RQ1 all come from parameter values that no declaration exercised. A practitioner should declare an operator for several representative parameters, including degenerate ones (order 0 and 1 filters, rate 1, default arguments), much as the SoP suites do. Generating parameters as well as signals is possible for some classes, e.g., stable filters from random poles, and is the most direct way to extend the method.

Complementary, not competing: The method and expected-value tests detected disjoint sets of faults, and together all of them. In practice they combine: a library keeps its regression tests, which pin values and parameter combinations, and adds relations, which need no values and explore inputs no one thought to write down. The cost is modest: one declaration per operator and a few seconds per run.

Soundness first: A method that reports faults in other people's code must not cry wolf. The validation of RQ3 is therefore part of the method, not an afterthought: proofs of the relations, an error model with a measured margin, references checked against 50-digit arithmetic, and generators audited against their own claims. The audit found a weakness in our own generators, and the campaign of RQ4 found three errors in our own declarations before any fault was reported upstream.

VIII. Conclusion

Signal processing code is hard to test because the correct output is unknown, the inputs that break it are specific, and floating-point results can only be compared within a tolerance that nobody derives: in the libraries we surveyed, tests compare with typed-in values, never generate their inputs, and use tolerances spread over 31 orders of magnitude.

We proposed a method that answers each problem from a single statement of what an operator is: metamorphic relations proved for four classes of DSP operators replace the expected output, signal generators target the inputs that break DSP code, and tolerances follow forward error bounds or a calibration that refuses to absorb faults. We evaluated it with a protocol that measures both its power and its soundness. On 166 generated faults it detects 90 %, against 96 % for the ported test suites of NumPy and SciPy, and the two are complementary: together

they detect 100 %. Its misses come from operator parameters that no declaration exercised; the misses of the suites, from code paths that their few expected values do not reach. On correct implementations it raised no false alarm in 2 000 runs, with references and generators that we audited.

Applied to widely used libraries, the method found faults in SciPy's `lfilter` and `sosfilt` on empty blocks with an initial state, the first being an independent rediscovery of a known issue, as well as a dropped delay in `tf2sos` and a clamping in torchaudio's `deemphasis`. We reported the empty-block faults to SciPy[2] and contributed a fix[3], the dropped delay of `tf2sos` to SciPy[4], and the clamping to torchaudio[5].

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AI usage disclosure

The authors conceived and directed this research: they defined the problem, the scope and the design of dsp-props, chose the experimental protocol, checked every result and submitted the bug reports to upstream projects. An AI coding assistant (Claude, Anthropic) was used under their direction to write code and experiment scripts and to assist with the writing. Every number in this article is produced by a script of the repository.

Conflict of interest

The authors declare no conflict of interest.

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APPENDIX A
PROOFS OF THE RELATIONS

Sequences have length N, indices are taken modulo N for the DFT, and $\mathcal{F}x[k] = \sum_{n=0}^{N-1} x[n]\,\omega^{kn}$ with $\omega = e^{-2\pi i/N}$. Every proposition of this appendix and the argument of Appendix B are also proved in Coq, with the Mathematical Components library, in the `proofs/` directory of the repository; its README maps each proposition to its Coq theorem and lists the modelling choices. The proofs use no axiom beyond the three classical axioms of MathComp Analysis, which only the specialisation to $\omega = e^{-2\pi i/N}$ needs.

Proposition 1 (Linearity)**.** *$\mathcal{F}$ is linear, and so is every LTI filter from zero state and every resampler of the form "upsample by L, filter, downsample by M".*

Proof. Each is a composition of linear maps: a matrix product for $\mathcal{F}$; zero insertion, convolution with h and decimation for the resampler; for a filter from zero state, y is the convolution $h * x$ truncated to N samples. □

Proposition 2 (Circular shift)**.** *If $x_s[n] = x[n-s]$, then $\mathcal{F}x_s[k] = \omega^{ks}\mathcal{F}x[k]$.*

Proof. $\sum_n x[n-s]\omega^{kn} = \sum_m x[m]\omega^{k(m+s)}$ with $m = n-s$ running over all residues. □

Proposition 3 (Parseval)**.** $\|\mathcal{F}x\|_2^2 = N\|x\|_2^2$.

Proof. $\mathcal{F} = \sqrt{N}\,U$ with U unitary, since $\sum_k \omega^{k(n-m)} = N\delta_{nm}$. □

A transform declared with another normalisation states its scale, which is checked instead of N.

Proposition 4 (Hermitian symmetry)**.** *If x is real, $\mathcal{F}x[N-k] = \overline{\mathcal{F}x[k]}$; in particular $\mathcal{F}x[0]$ and, for even N, $\mathcal{F}x[N/2]$ are real.*

Proof. $\omega^{(N-k)n} = \overline{\omega^{kn}}$ and x is real. For $k = 0$ and $k = N/2$, the bin is its own conjugate. □

Proposition 5 (Inversion and convolution)**.** *$\mathcal{F}^{-1}\mathcal{F}x = x$, and $\mathcal{F}^{-1}(\mathcal{F}x \cdot \mathcal{F}y) = x \circledast y$, the circular convolution.*

Proof. The first follows from $\mathcal{F}^{-1} = \frac{1}{N}\overline{\mathcal{F}}$ and Parseval's orthogonality. For the second, $\mathcal{F}(x \circledast y)[k] = \sum_n \sum_m x[m]y[n-m]\omega^{kn} = \mathcal{F}x[k]\,\mathcal{F}y[k]$ by the shift proposition. □

dsp-props computes $x \circledast y$ directly in $O(N^2)$, independently of the operator under test.

Proposition 6 (Time invariance and causality on finite sequences)**.** *Let f be a causal LTI filter applied from zero state, $f(x)[n] = \sum_{m \le n} h[n-m]x[m]$. Then (i) $f([0_s, x]) = [0_s, f(x)]$ on $N+s$ samples, and (ii) if x and x' agree on $[0,k)$, then $f(x)$ and $f(x')$ agree on $[0,k)$.*

Proof. (i) For $n < s$ the sum only involves zeros; for $n \ge s$, substitute $m' = m - s$. (ii) $f(x)[n]$ for $n < k$ only involves $x[m]$, $m \le n < k$. □

For a non-causal filter (e.g., a centred FIR), (i) holds after discarding the first s output samples, which is what the library checks.

Proposition 7 (Transfer function)**.** *For a causal filter with transfer function $H = B/A$ whose impulse response h is absolutely summable, and k with $A(e^{2\pi ik/L}) \neq 0$, the length-L DFT of $(h[0], \ldots, h[L-1])$ equals $H(e^{2\pi ik/L})$ up to the aliasing term $\sum_{n \ge L} h[n]\omega_L^{kn}$, whose modulus is at most $\sum_{n \ge L} |h[n]|$.*

Proof. $H(e^{i\theta}) = \sum_{n \ge 0} h[n]e^{-i\theta n}$ converges absolutely for a stable filter; split the sum at L. □

The library chooses L so that $r^L < 10^{-17}$, with r the largest pole modulus, and adds the tail to the tolerance.

Proposition 8 (Bounded gain)**.** *For an LTI filter with $\|h\|_1 \le G$, every input with $|x[n]| \le 1$ gives $|y[n]| \le G$, and the input $x[m] = \operatorname{sign}(h[N-1-m])$ attains $\sum_{n<N} |h[n]|$ at $n = N-1$.*

Proof. $|y[n]| \le \sum_m |h[n-m]||x[m]| \le \|h\|_1$; with the given input, every term of $y[N-1] = \sum_m h[N-1-m]x[m]$ is $|h[N-1-m]|$. □

Proposition 9 (Periodic time invariance)**.** *Let $L, M \ge 1$ and $f = {\downarrow_M} \circ (h * \cdot) \circ {\uparrow_L}$, the filter being applied from zero state. Delaying the input by Ms samples delays the output by Ls samples.*

Proof. $\uparrow_L$ turns a delay of Ms into a delay of LMs; the filter preserves it; $\downarrow_M$ turns a delay of LMs, a multiple of M, into a delay of Ls. □

The output length $\lceil NL/M \rceil$, the DC gain and the alias rejection are declared specifications rather than theorems; their tolerances (ripple, stop-band level) are part of the declaration.

Proposition 10 (Block invariance)**.** *Let a streaming processor be a state machine $(q_{t+1}, y_t) = \phi(q_t, x_t)$ applied sample by sample, and let `process` on a block apply ϕ to its samples in order, starting from the current state and storing the final one. Then for any split $x = x^{(1)} \cdots x^{(J)}$, including empty blocks, concat_j `process`$(x^{(j)})$ = `process`(x) from the same initial state.*

Proof. By induction on J: an empty block leaves the state unchanged and outputs nothing, and processing $x^{(1)}$ then $x^{(2)}$ applies ϕ to the samples of $x^{(1)}x^{(2)}$ in order. □

Block-based algorithms such as overlap-add are not sample-by-sample machines, but a correct one computes the same function, which is what the relation checks.

APPENDIX B
ERROR BOUND OF A LINEAR RECURSION

Normalise $a_0 = 1$. The direct recursion computes $y[n] = \sum_{j=0}^{q} b_j x[n-j] - \sum_{j=1}^{p} a_j y[n-j]$ with $k = p + q + 1$ products, one per coefficient other than a_0. In floating point, with the standard model [23], the computed $\hat{y}$ satisfies $\hat{y}[n] = \sum_j b_j x[n-j] - \sum_j a_j \hat{y}[n-j] + \delta[n]$ with $|\delta[n]| \le \gamma_k \big(\sum_j |b_j||x[n-j]| + \sum_j |a_j||\hat{y}[n-j]| \big)$, $\gamma_k = k\varepsilon/(1-k\varepsilon)$. The Coq proof takes this bound on δ as a hypothesis; everything after it is proved. The error $e = \hat{y} - y$ therefore obeys $A(z)e = \delta$, i.e., $e = h_A * \delta$. By Young's inequality $\|e\|_2 \le \|h_A\|_1 \|\delta\|_2$, and $\|\delta\|_2 \le \gamma_k (\|b\|_1 \|x\|_2 + \|a_{1:}\|_1 \|\hat{y}\|_2)$. Finally $\|\hat{y}\|_2 \le \|h_{B/A}\|_1 \|x\|_2 + \|e\|_2$, which gives (2) with $c = 1$ to first order in ε, with $\gamma_k \approx k\varepsilon$. Rigorously, if $k\varepsilon \le 1/2$ and $2k\varepsilon \|h_A\|_1 \|a_{1:}\|_1 \le 1/2$, the bound holds with $c = 4$ for one computation. The library uses $c = 4$ for the difference of the two computations compared in a relation, twice the first-order requirement $c = 2$, and the validation of Section VI-C confirms that the bound is never reached on correct code.